ANALYSIS AND SYSTEMS OF BIG DATA | COURSE PROJECT

Shadow Squad

Who replaces whom? Ranking, pricing and stress-testing replacements across Europe's top five football leagues

End-Semester Report | November 2026

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At a glance

Big Data · End-semester summary

Question. The mid-sem asked which young players will move up. The end-sem asks what a club asks: *a starter is leaving, who should replace him?* The system is tested against what clubs really did: 593 elite replacement events and 1,941 summer signings by any club in the five leagues, 2017-18 to 2024-25. Every model that ranks a season was fitted on earlier seasons only.

11,792 player-seasons, 8 seasons, 5 leagues; three sources joined in a Spark + Parquet lake	**99.7%** of 900-minute players linked from FBref to Transfermarkt ids	**13% vs 4%** recall@10, engine against market value, on 1,029 unseen signings	**14% vs 0%** recall@10 on the 301 signings below the median fee, engine against market value
+176% mean two-year return of the 25 most under-priced players each summer; a random 25 returns +22%	**+1.8 pts** change in share of league minutes from moving to an elite club, against a matched twin (95% CI -4.3 to +7.9)	**30%** of near-miss players can pad cheap stats to enter the top 10; the hardened model brings this to zero	**8 + 7** new Spark algorithms (one per task), plus seven techniques beyond the course

Most important findings (all computed; details in sections 6 and 7):

- **The engine finds real signings.** On 1,029 unseen signings it puts the bought player in the top 10 for **13%** of events, against 4% for market value, 7% for the best young performer and 6% for a random order ($p < 0.001$ against market value).
- **It earns its keep on cheap signings.** Over all signings below the median fee (EUR 14m) the engine reaches 14% recall@10 and market value 0.3% ($p < 0.001$); above the median the gap narrows to 19% against 13%. The dearer the signing, the more market value already knows.
- **The market misprices, and it can be measured.** A portfolio of the 25 players priced furthest below what their profile predicts returned **+176%** in two years (+149% on the money spent), against +22% for a random 25 ($p < 0.001$). The league discount is real but is not a bargain: it does not close.
- **Moving to an elite club neither helps nor hurts a young player on average.** Against a twin who stayed, minutes change by +1.8 points and style-neutral output by -0.05 index points (0-100 scale); both intervals include zero (142 matched pairs).
- **The scout can be gamed, cheaply, and fixed for free.** 30% of players ranked 11 to 20 enter the top 10 by raising cheap stats (median: 0.09 more shots per 90). A model that ignores those stats closes the hole with recall@10 at 36% against 31%.
- **What did not work.** On elite replacements overall the engine (31% recall@10) does not beat market value (35%) or the hazard score (35%). Style translation is no better than raw numbers once refitted walk-forward, ALS club taste loses to plain popularity, and the engine's own top-25 portfolio is no better than random.

WHAT CHANGED SINCE THE FIRST END-SEM DRAFT

Market value is now the last valuation before the window opens, so it cannot react to the signing it predicts. Style elasticities, standardisation, archetypes and outlier scores are refitted at every season cut-off on past seasons only. The 2024-25 season was added, and a third matching pass cut unmatched feeder-seasons to 13 of 3,154. Numbers from the first draft are replaced by the ones above.

Contents

1	What changed since mid-sem	6	Results: who gets signed
2	Data lake and PySpark pipeline	7	Beyond the syllabus
3	New algorithms, one per task	8	Model predictions and case studies
4	Techniques beyond the course	9	Limitations and conclusion
5	Style translation on real movers		

1 What changed since mid-sem

Problem · Labels · Leakage control

The mid-sem report (included unchanged before this part) found that no method clearly beat ranking the best young performers, and that its cosine shortlist was worse than that baseline. Three things were limiting it.

The question. A club does not buy "a player who will move up". It buys a replacement for a starter who left, or fills a position. The end-sem system takes a club, a season and a departing starter (or a position), and returns a ranked list. The test is whether the player the club really signed is near the top.

The labels. Mid-sem outcomes were inferred from the club a player's name appeared next to one or two seasons later. They now come from Transfermarkt transfer records. On the 1,966 feeder-seasons with both labels, the two agree on "moved up" for 91.7%. The old rule called 132 moves with no matching permanent transfer, mostly loan spells and loan returns; on the same rows the move-up rate falls from 26.5% to 21.4%. The new label also covers 2022-23 and 2023-24.

The data. Transfermarkt (*davidcariboo/player-scores*) adds transfers, fees, market values, fine positions and a stable player id. FBref removed its advanced statistics in January 2026; a scrape taken before that (*emrey3lmaz*) supplies 2024-25. The two FBref sources agree on the shared 2023-24 season: the lowest per-metric correlation over 1,600 players is 1.000.

The mid-sem notebook was re-run unchanged first. Its minutes column is named *avg_mins_per_match* but holds season totals (maximum 3420), so no fix was needed. Its baseline on the refined labels: feeder base rate 26.7%; best walk-forward model 46 hits in 75 picks; best-composite baseline 31 in 75.

Leakage control

Component	How it is kept out of the future
Market value	last valuation before 1 June after season t; the window used for the events opens that day
Style elasticities, league offsets	refitted for every season on seasons up to it; the values correlate 0.996 with the mid-sem all-season fit
Standardisation (z-scores)	mean and SD from seasons up to the row's own
Archetypes, outlier scores	number of clusters chosen on seasons up to 2019-20; clustering refitted at each cut-off; neighbours taken from past seasons only
PageRank, ALS taste	graph and signing matrix built from transfers dated before June of t+1
Hazard, ranker	trained on seasons or events up to t-1

Table 1. Every input a ranking of season t uses, and the data it was fitted on.

2 Data lake and PySpark pipeline

Spark · Parquet · Entity resolution



Figure 1. The pipeline. Every stage after the first reads and writes Parquet under lake/.

All sources are written as Parquet partitioned by season and league: 21,097 raw FBref player-club rows, 11,792 player-seasons with 900 minutes or more, 780 Understat club-seasons, 168,248 transfers, 615,190 valuations and 50,149 Transfermarkt players. League tables are rebuilt from match results and agree with the mid-sem table on 99.1% of club-seasons. Writing the lake takes 20 s in local mode, and a query for one season and league reads one partition.

Entity resolution. FBref has no player id. Pass 1 learns the club alias map from the data (join on season, league, birth year and exact name key, then a majority vote): all 149 clubs map one to one. Pass 2 blocks on season and club and compares accent-free, token-sorted name keys with Spark *levenshtein* and a Jaro-Winkler pandas UDF, plus birth year and position group, keeping mutual best pairs. Pass 3 handles nicknames that share no letters (FBref "Jonny Castro" is Transfermarkt "Jonny Otto"): same club and season, same birth year and position group, league minutes within 10%, unique on both sides. It added 34 players. In all, 98.3% of outfield player-club-seasons are linked and 99.7% of those with 900 minutes; 13 of 3,154 feeder-seasons stay unmatched and are listed in *outputs/unmatched_feeders.csv*.

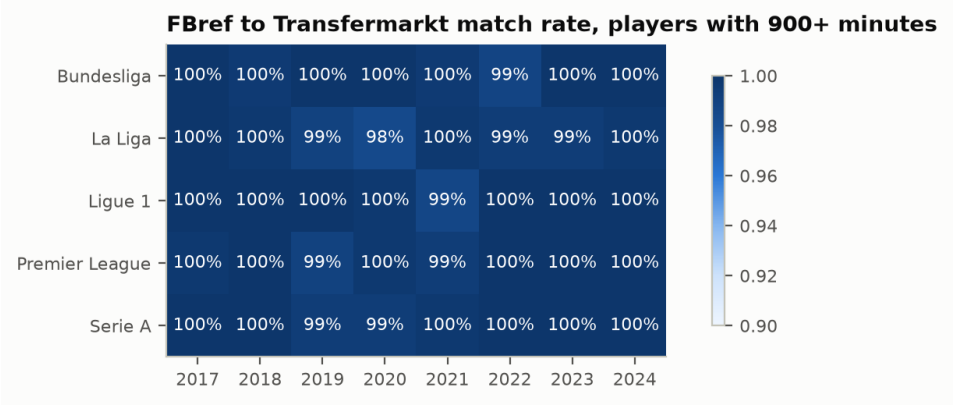


Figure 2. FBref to Transfermarkt match rate per league and season, players with 900 minutes or more.

Transfers. Loans are found by walking each player's moves in date order: a move that undoes the previous one within 18 months closes a loan, and a loan made permanent counts from the day it started (23,100 loan departures removed). The dump has no transfer record at all for some players, so their moves are read from league minutes (1,417 moves, 750 players). After that 96.0% of feeder-seasons have a transfer history.

Two event sets. An *elite replacement* is an elite club whose starter (1,350 minutes or more) plays under 450 league minutes for it the next season, and which signs a player of the same fine position that summer who then plays at least 450 minutes: 593 events from 479 signings. Only 24% of those signings were inside the candidate pool (age 26 or under, 900 minutes, non-elite club in the five leagues); 25% came from clubs outside the five leagues and the rest were older, at another elite club, or had played too little. The engine can only be right about the 128 events whose signing was in the pool. The second set is *every summer signing* by a five-league club of a player with 900 minutes in the five leagues the season before: 1,941 signings in 1,657 club-season-position queries.

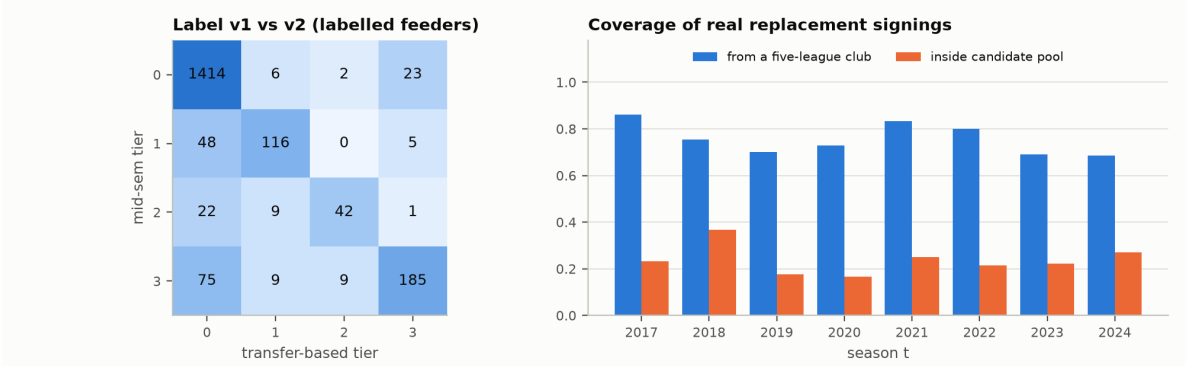


Figure 3. Left: mid-sem label against the transfer label. Right: share of real replacement signings from a five-league club, and inside the candidate pool.

3 New algorithms, one per task

PySpark · 8 mining tasks

None of these was used before the mid-sem. Each runs in Spark (local mode) with its physical plan shown once in notebook 03.

Task	Algorithm (Spark)	Result	Time
Similarity search	Euclidean LSH, MinHash LSH	95% of true top-25 neighbours kept while comparing 88% of pairs	0.5 s
Sequential patterns	PrefixSpan	461 frequent career sequences; flagged players move up 27.4% against 24.9% out of time	1.2 s
Clustering	Bisecting k-means per fine position	22 archetypes, mean silhouette 0.19; Cramer's V with moving up 0.19	120 s
Link analysis	GraphFrames PageRank; power-iteration clustering	pipeline rank AUC 0.582 (sales count 0.582); communities with modularity 0.33	24 s
Recommendation	ALS, implicit feedback	MRR 0.633 against 0.668 for popularity: ALS loses	15 s
Classification	LinearSVC, multilayer perceptron	LinearSVC AUC 0.690, MLP 0.604 (mid-sem best 0.681)	161 s
Outlier detection	k-NN distance score (self-join)	flagged "hidden gems" move up 16.7% against 22.8%: the flag does not work	2.8 s
Stream mining	Structured Streaming, Count-Min Sketch	live team dominance reaches r = 0.9 with the season value after about 12 matches	13 s

Table 2. The eight Spark mining tasks, each with its own metric.

Similarity search (LSH). Random-projection LSH hashes vectors so that close ones tend to share a bucket, and only bucket-mates are compared. It generates candidates for each replacement event from style-neutral vectors; MinHash on top-quartile trait sets is a second view. The pool is small (about 95 same-position players per event), so LSH saves little here: the chosen setting still compares 88% of pairs. The same code would matter on a pool of every league in the world.

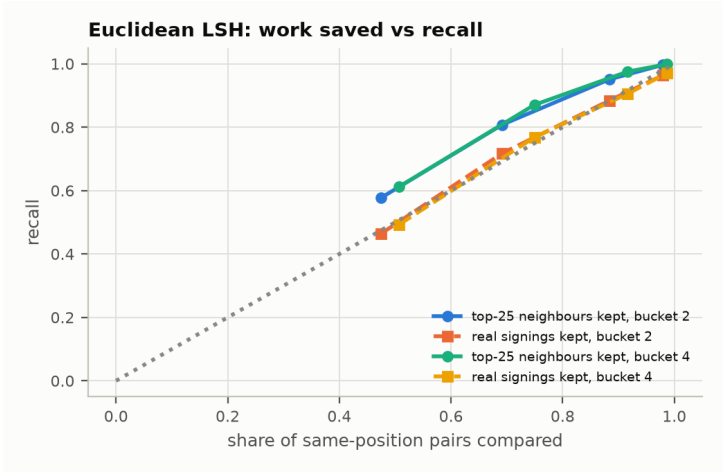


Figure 4. LSH: share of pairs compared against recall of the exact top-25 neighbours and of the real signings.

Sequential patterns (PrefixSpan). A career is an ordered list of seasons, each an itemset of club tier, league, role and form band. PrefixSpan mines frequent orders on feeder-seasons up to 2019-20. The strongest multi-season pattern is {club=mid} -> {club=mid, form=mid, role=starter} (lift 2.04 in training, 1.46 later). Out of time the flagged group moves up at 1.10 times the base rate (Fisher p <0.001): real, and small.

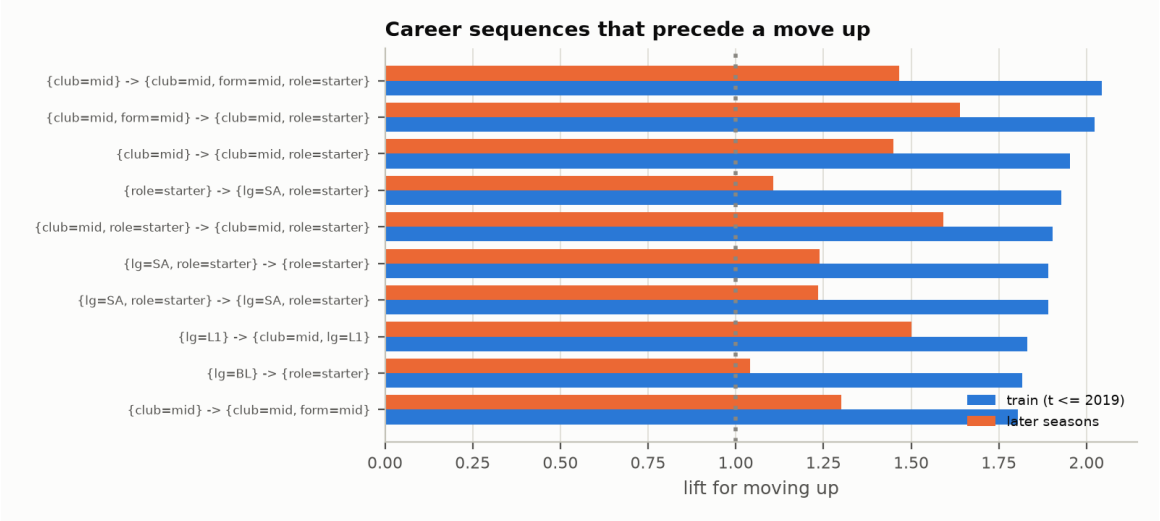


Figure 5. Career sequences with the highest lift for moving up, in training and in later seasons.

Clustering (bisecting k-means). Bisecting k-means splits the worst cluster in two, repeatedly. It runs inside each of the seven Transfermarkt fine positions, which the mid-sem could not do with three coarse groups, and is refitted at every season cut-off. It finds 22 archetypes; archetype and moving up are associated (chi-square p <0.001, Cramer's V 0.19). Creative archetypes are bought more often than defensive ones at every position.

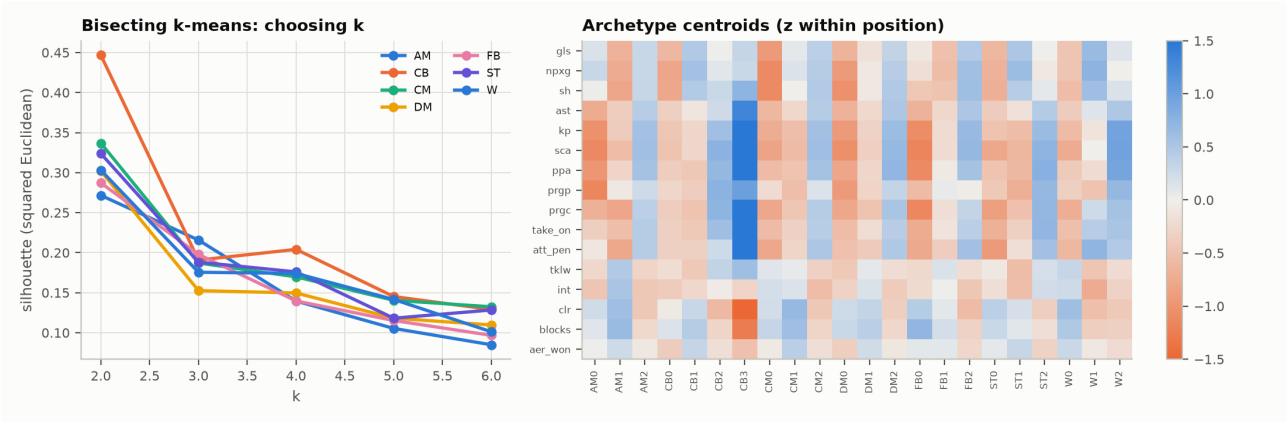


Figure 6. Silhouette by number of clusters for each position, and the archetype centroids.

Link analysis (PageRank, PIC). Each permanent transfer is an edge (62,535 in all). With edges pointing from buyer to seller, PageRank is high for clubs whose players are bought by important clubs. Academies and one-off amateur clubs were removed first, because the reversed graph rewards them. The top of the list is Red Star, Salzburg, Inter Milan, Olympiacos, Boca Juniors. For feeders, the club's pipeline rank alone gives AUC 0.582, the same as counting the club's past sales (0.582). Power-iteration clustering on 1,002 clubs gives transfer communities with modularity 0.33; the iteration count is chosen by modularity, because too many iterations collapse every club into one group.

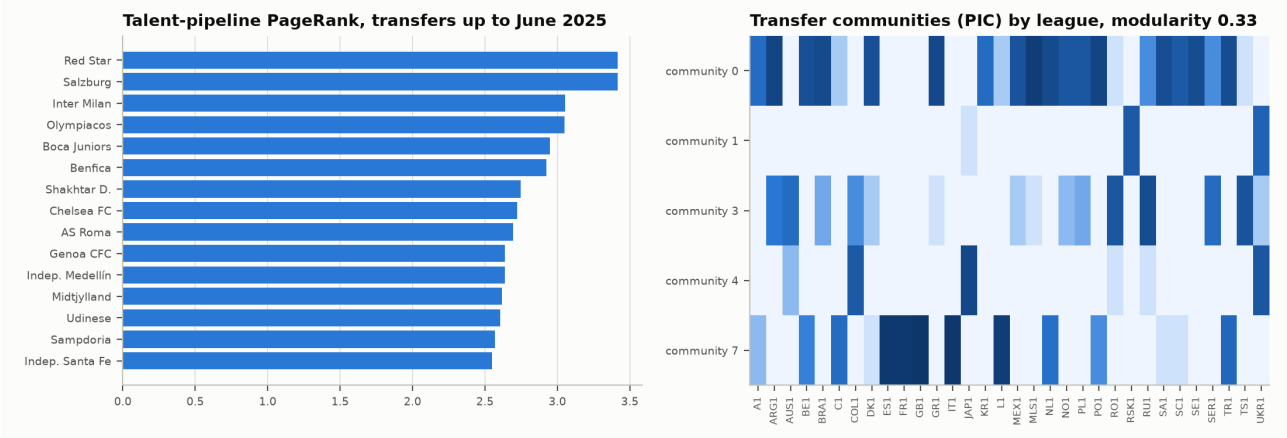


Figure 7. Left: clubs with the highest talent-pipeline PageRank. Right: transfer communities by league.

Recommendation (ALS). ALS factorises the club by archetype matrix of past signings. It did not work: on 1,551 later signings it ranks the bought archetype worse than league-wide popularity (MRR 0.633 against 0.668; 95% interval of the difference -0.052 to -0.019). The training matrix has 560 signings across 144 clubs, too few to learn a club's taste. The score stays in the engine so the ablation can measure it.

Classification (LinearSVC, MLP). Same question and walk-forward split as the mid-sem classifiers. LinearSVC reaches AUC 0.690 with 41 hits in 75 picks, level with the mid-sem logistic regression (0.681, 46 hits). The perceptron overfits the small training set (AUC 0.604). On the transfer label over five test seasons LinearSVC gets AUC 0.681 and 59 hits in 125.

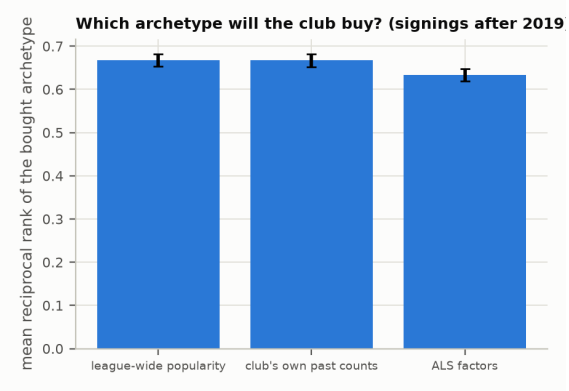


Figure 8. Mean reciprocal rank of the archetype a club actually bought.

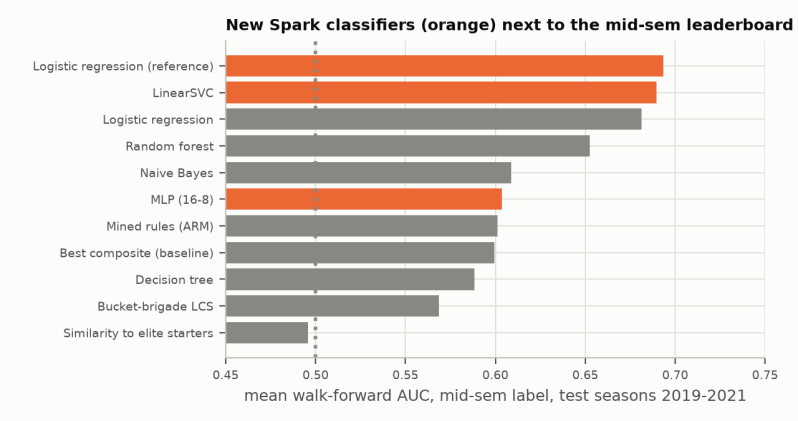


Figure 9. New Spark classifiers next to the mid-sem leaderboard (same label, same test seasons).

Outlier detection (k-NN distance). The score is the mean distance to the ten nearest neighbours inside the archetype, with neighbours taken from past seasons only. Unusual players are bought slightly *less* often (Spearman -0.081, p <0.001), and the "hidden gem" flag (top decile of the score, above-median composite index) marks 114 feeders who moved up at 16.7% against 22.8% for the rest. An earlier fit on all seasons had suggested the opposite; it did not survive the walk-forward refit.

Stream mining. The 2023-24 Understat matches are replayed one week per micro-batch (39 batches). A stateful aggregation keeps each team's running dominance; after five weeks it correlates 0.80 with the end-of-season value and passes 0.9 at about 12 matches per team. A Count-Min Sketch counts deep completions per team in fixed memory: at width 200 its mean relative error is 0.0%, and at width 20 it over-counts by 2.5 times the true count, which is what the width buys.

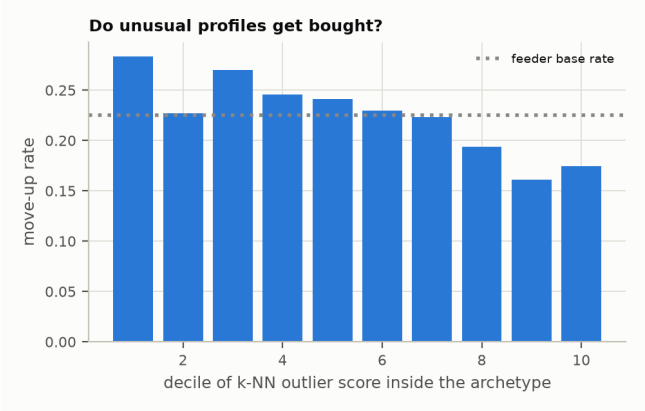


Figure 10. Move-up rate by decile of the k-NN outlier score.

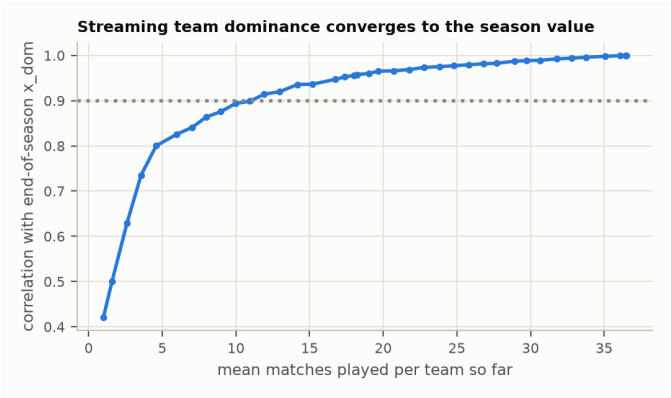


Figure 11. Streaming team dominance against the end-of-season value.

4 Techniques beyond the course

Survival · Ranking · Conformal · SHAP · DTW

Survival analysis. Feeder-seasons become person-period rows (2,947 rows, 382 events), which lets the latest seasons be used. The discrete-time hazard is a Spark logistic regression refitted each season on past rows only. Its walk-forward AUC is 0.686, and predicted and observed rates differ by 2.1% on average across deciles. On 1,037 later entrants its concordance index is 0.676; a Cox proportional-hazards model (*lifelines*) fitted as a check gets 0.654.

Learning to rank. LightGBM LambdaMART with one query per event, against a pointwise Spark GBTClassifier. Section 6 has the results.

Conformal prediction. Split-conformal turns a score into a list with a stated coverage. For movers: a hazard list that must catch 90% of next season's movers holds 81% of all feeders and caught 93.4% on later seasons; a 50% promise needs 29% and caught 53.3%. For replacement events (calibrated on 26 events, checked on 65): a list that contains the real signing 90% of the time needs about 88 players and covered 96.9%; a 50% promise needs about 20 and covered 44.6%. The guarantees hold, and they show that a short list cannot promise much.

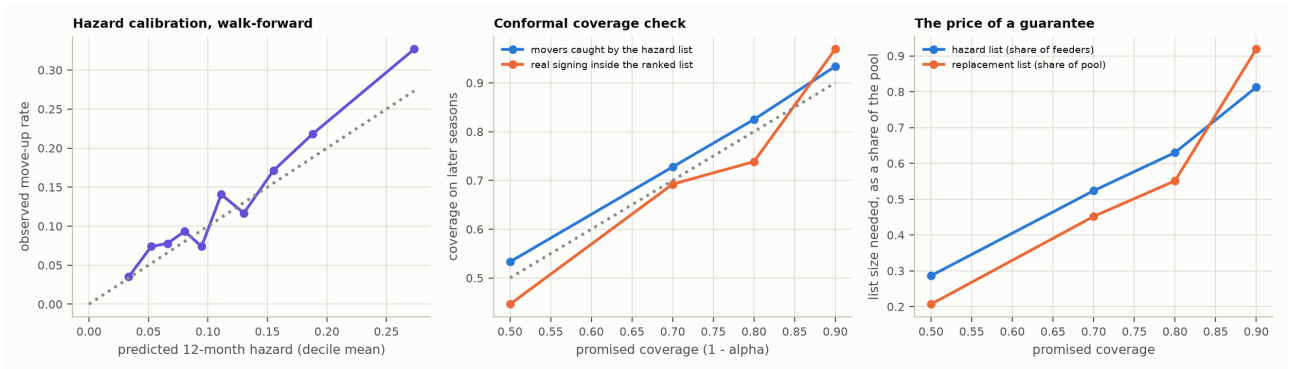


Figure 12. Hazard calibration, conformal coverage on later seasons, and the list size a guarantee costs.

SHAP (TreeExplainer on the ranker) explains each candidate's rank in section 8. **Dynamic time warping** is compared with the simple trajectory similarity: the two correlate 0.45 and put the real signing in the top 10 about equally often (16% and 17%), so DTW adds nothing here. Only 15% of pairs could be age-aligned, because the starter's seasons at the candidate's age are usually before 2017. Causal inference, hedonic pricing and adversarial testing are in section 7.

5 Style translation on real movers

Domain model · Validation

The mid-sem removed team style from a player's numbers. Translation runs it the other way, into the buying club:

$$\text{stat in club } C = \text{neutral stat} \times \exp\left(\Sigma \text{ supported } \beta(\text{position, metric, term}) \times \text{style term of } C\right) \times \text{league factor of } C$$

with elasticities and league offsets refitted at each season cut-off. The check uses 1,319 players who changed club between consecutive seasons (557 across leagues). Their per-90 numbers at the new club are predicted from (a) last season's raw numbers, (b) style-neutral numbers, and (c) neutral numbers translated into the new club, using the club's style from the season before the move.

Mean absolute error, in standard deviations of each metric: (a) 0.495, (b) 0.492, (c) 0.494. The three are the same within noise: the 95% interval for (c) minus (a) is -0.004 to +0.003, and (c) is better in 30 of 48 position-by-metric cells. Using the new club's true style that season does not help either (0.491).

FINDING

Translation does not predict a mover's numbers better than leaving them unchanged. The first draft reported a small gain; it came from elasticities fitted on all seasons and disappears when they are fitted on past seasons only. Translation is therefore kept as a display feature (the radar in section 8) and as one ranking score whose value the ablation measures, not as a forecasting tool.

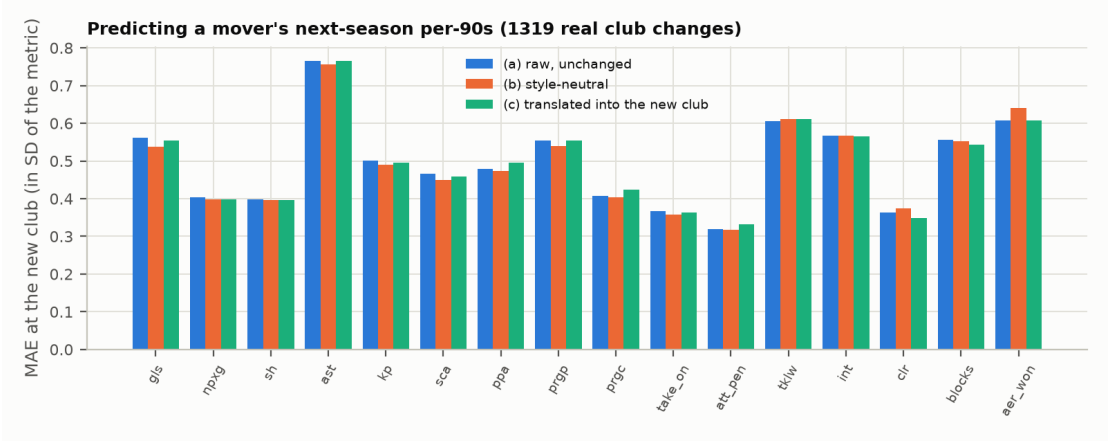


Figure 13. Error in predicting a mover's next-season per-90s, by metric.

6 Results: who gets signed

Evaluation · Baselines · Uncertainty

For each query the engine scores every candidate on nine things: translated similarity to the reference profile, trajectory similarity, the selling club's pipeline PageRank, the buying club's ALS taste, the survival hazard, market value (level, and gap to the reference), age and composite index. A LambdaMART ranker combines them. Queries of season Y are ranked by a model that saw only signings up to Y-1.

6.1 Every summer signing

This is the large test: 1,029 unseen club-season-position queries from 2020-21 to 2024-25, about 229 candidates each. The reference profile is what the club already fields at that position.

Method	Median rank	Recall@5	Recall@10 [95% CI]	Recall@25	Recall@50	p vs market value
Random order of the pool	90	3%	6% [4, 7]	14%	29%	

Method	Median rank	Recall@5	Recall@10 [95% CI]	Recall@25	Recall@50	p vs market value
Highest market value	90	2%	4% [3, 6]	12%	27%	
Raw cosine similarity to the club profile	90	4%	6% [5, 8]	16%	29%	0.04
Best young performer (composite index)	90	4%	7% [5, 8]	15%	30%	0.01
Single score: hazard	77	4%	8% [7, 10]	19%	35%	<0.001
Shadow Squad	**53**	**8%**	**13% [11, 15]**	**28%**	**48%**	**<0.001**

Table 3. All signings: recall@K and median rank of the bought player on 1,029 unseen queries. Intervals: bootstrap over queries. Last column: paired permutation test on recall@10.

FINDING

The engine puts the bought player in the top 10 for 13% of signings and in the top 25 for 28%. Market value manages 4% and 12%, below the best young performer (7%) and barely different from random (6%): most clubs cannot afford the most valuable players, so value says little about whom they buy. For clubs outside the elite the engine reaches 11% and market value 0%.

6.2 Elite replacements

The original task: 91 unseen events whose real signing was in the candidate pool, about 98 candidates each. The reference profile is the departing starter.

Method	Median rank	Recall@5	Recall@10 [95% CI]	Recall@25	Recall@50	p vs market value
Random order of the pool	44.0	6%	12% [5, 19]	30%	57%	
Raw cosine similarity to the starter	30.0	13%	23% [14, 32]	42%	65%	0.07
Best young performer (composite index)	30.0	13%	26% [18, 35]	45%	67%	0.18
Highest market value	15.5	23%	35% [25, 45]	64%	84%	
Single score: hazard	16.0	19%	35% [25, 45]	64%	84%	1.00
Single score: translated similarity	32.0	14%	24% [15, 33]	40%	64%	0.11
Shadow Squad fusion with Spark GBT (pointwise)	25.0	5%	18% [10, 25]	51%	71%	0.02
Shadow Squad, replacement events only	21.0	10%	26% [18, 35]	56%	79%	0.23
Shadow Squad (trained on all signings)	**18.0**	**22%**	**31% [21, 41]**	**60%**	**78%**	**0.60**
Ablation: no translation	19.0	24%	35% [25, 45]	57%	76%	1.00
Ablation: no trajectory	18.0	23%	35% [25, 45]	63%	77%	1.00
Ablation: no PageRank	18.0	20%	32% [22, 42]	62%	78%	0.72
Ablation: no ALS taste	18.0	15%	32% [22, 42]	64%	73%	0.71

Table 4. Elite replacements: 91 unseen in-pool events. The first Shadow Squad row learns from replacement events only; the bold row learns from all signings and is the engine used everywhere else.

What worked. Training on all signings instead of the few replacement events lifts the engine from 26% to 31% recall@10 and from a median rank of 21 to 18. LambdaMART beats the pointwise Spark GBT (18%). The engine is well above random (12%) and the mid-sem cosine method (23%).

What did not. On this task the engine does not beat two single scores. Market value reaches 35% recall@10 with a median rank of 15.5, and the hazard alone 35%. The difference between the engine and market value is inside the

noise (recall@10 difference -16 to +8 points, p 0.60), and so is its lead over the best young performer (-8 to +16 points). Removing one score at a time moves recall@10 between 32% and 35%; every interval overlaps the full model's, so no single score can be called necessary on 91 events. The engine is the average of ten LambdaMART models, each trained on its own 80% subsample. One such model alone gives between 27% and 40% recall@10 depending on its seed (mean 32%), and between 16% and 38% on the cheaper half, so a few points either way on this task are noise.

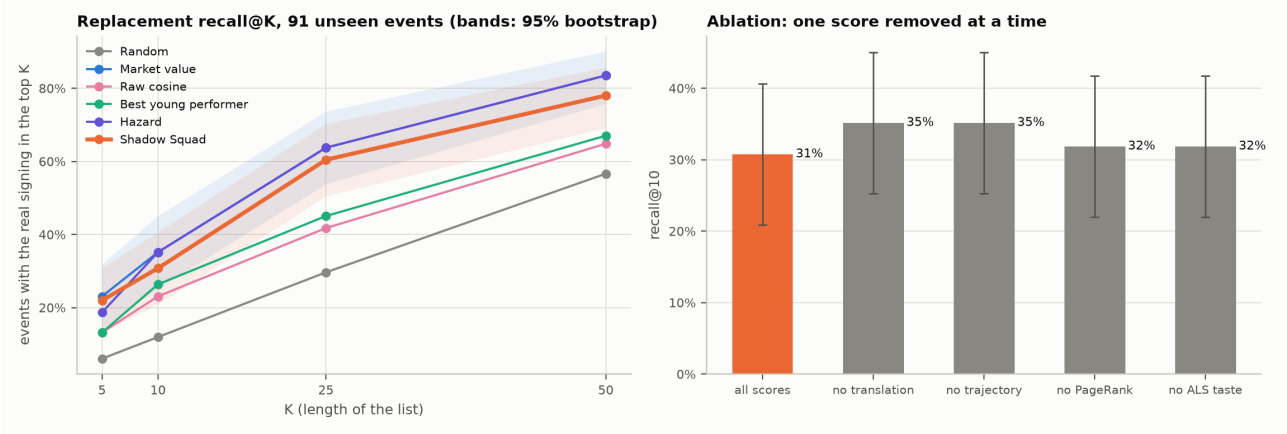


Figure 14. Elite replacements. Left: recall@K with 95% bootstrap bands for the engine and market value. Right: recall@10 with one score removed at a time.

6.3 Cheap against expensive signings

Why does market value do well on elite replacements and badly everywhere else? Split the events at the median fee of the real signing. Over all signings (median EUR 14m) the engine is far ahead below the median, and the gap narrows above it. Among elite replacements there are only 37 events in each half: in the cheaper half (below EUR 24m) the engine reaches 24% and market value 11% (p 0.23); in the dearer half it is 41% against 65% (p 0.04). The direction matches the large set, but on this few events the cheaper-half difference is not significant.

Events	n	Engine R@10	Market value R@10	Difference, points [95% CI]	p	Engine R@25	Market R@25
Elite replacements: below median fee	37	24%	11%	+14 [-3 to +32]	0.23	54%	46%
Elite replacements: above median fee	37	41%	65%	-24 [-43 to -5]	0.04	65%	89%
Elite replacements: free or undisclosed	17	24%	24%	+0 [-29 to +29]	1.00	65%	47%
All signings: below median fee	301	14%	0%	+13 [+10 to +17]	<0.001	29%	1%
All signings: above median fee	300	19%	13%	+6 [+1 to +12]	0.04	35%	30%
All signings: free or undisclosed	428	9%	1%	+8 [+5 to +11]	<0.001	22%	7%

Table 5. Recall by fee tier of the real signing. Permutation p-values are paired, two-sided, on recall@10.

FINDING

On the large set the hypothesis holds: below the median fee the engine finds 14% of signings in its top 10 and market value 0.3%; above it, 19% against 13% (p 0.04). For free and undisclosed moves it is 9% against 1.4%. A club that wants the obvious star does not need a model. A club shopping below EUR 14m does. The elite-replacement split is too small to settle the question on its own.

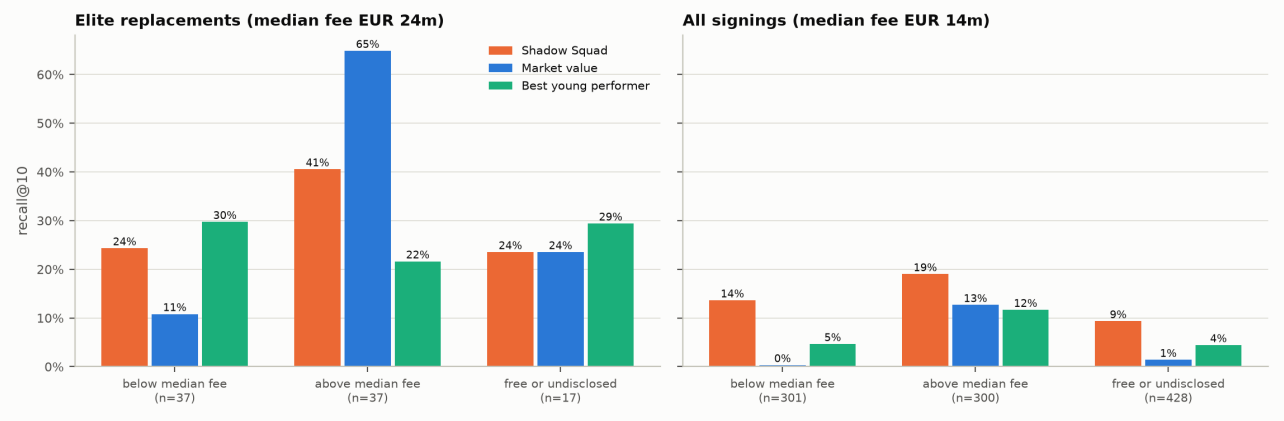


Figure 15. Recall@10 by fee tier of the real signing, for elite replacements and for all signings.

7 Beyond the syllabus

Pricing · Causal inference · Adversarial ML

Each question ends in one sentence, backed by a number with an interval. Null results are reported as they are.

7A. Scouting as an investment portfolio: where does the market misprice?

Each summer from 2020 to 2023 a portfolio of 25 pool players is bought at Transfermarkt value (last valuation before the window opens). The return is the realised fee if he was sold for a fee within 24 months, otherwise his value 24 months later. Five portfolios are compared with 10,000 random portfolios of the same size. Section 6 showed that market value matches the engine at the expensive end, so the useful question is where the market is wrong. A hedonic model answers it: regress log market value on the style-neutral profile, age, league, minutes and club tier (R^2 0.71, 11,745 player-seasons, standard errors clustered by player). The residual is how far a player is priced from what his profile predicts. The "under-priced" portfolio buys the 25 most negative residuals, with the model fitted on seasons up to that summer.

Portfolio (top 25 per summer)	Mean return [95% CI]	Return on money spent	Median	Hit rate	Sharpe	p vs random	Sold for a fee	Return on those sales
Under-priced (hedonic)	**+176% [+103 to +276]**	**+149%**	**+37%**	**63%**	**0.39**	**<0.001**	**16**	**+140%**
Best young performer	+28% [+7 to +56]	+13%	+0%	47%	0.24	0.27	43	+25%
Hazard	+22% [+7 to +41]	+16%	+0%	45%	0.26	0.43	63	+30%
Shadow Squad	+18% [+2 to +36]	+13%	+0%	41%	0.21	0.57	49	+13%
Market value	-1% [-9 to +8]	-1%	-8%	39%	-0.01	0.97	49	+12%
Random	+22% [-1 to +58]	+7%	-10%	39%	0.15			

Table 6. Two-year returns of top-25 portfolios, 2020-2023 (100 picks each). Hit rate: share of picks whose value rose. Sharpe: mean over standard deviation of player returns. Last column: realised fees only, the robustness check.

HEADLINE

A club buying the 25 most under-priced young players each summer would have gained +176% per player over two years (+149% on the money spent), against +28% for buying the best young performers, +18% for the engine's top 25 and +22% for a random 25.

Where the market is wrong. Like for like, a player in La Liga is valued 48% below the same profile in the Premier League, and one in Ligue 1 64% below. That discount is not a bargain: in the return model (next-24-month log return on the same attributes plus today's value) every non-Premier-League coefficient is negative, so the gap widens. What the market does under-price is output it has already seen but not yet paid for: sca, gls, npxg have the largest positive return coefficients (sca +0.048, p 0.003; gls +0.043, p <0.001; npxg +0.043, p <0.001), and playing time has the largest of all (+0.68 per full season of minutes). Value also reverts: the coefficient on today's log value is -0.34. Across the pool, the most under-priced fifth returned +85% and the most over-priced fifth -17% (Spearman -0.26, p <0.001).

The engine as an investor. Its top 25 returned +18%, no better than random (p 0.57). It is trained to find whom elite clubs sign, and those players are already priced. Buying the 25 most valuable players lost money (-1%).

Limitations. Transfermarkt values are crowd estimates and react to transfers, so value growth is partly circular; cheap players also have more room to grow, and the under-priced portfolio spends only EUR 169m against EUR 4447m for the market-value portfolio. On realised fees alone the under-priced picks that were sold (16 of 100) returned +140%, so the result does not rest on crowd values, but few were sold. Four summers is a short record.

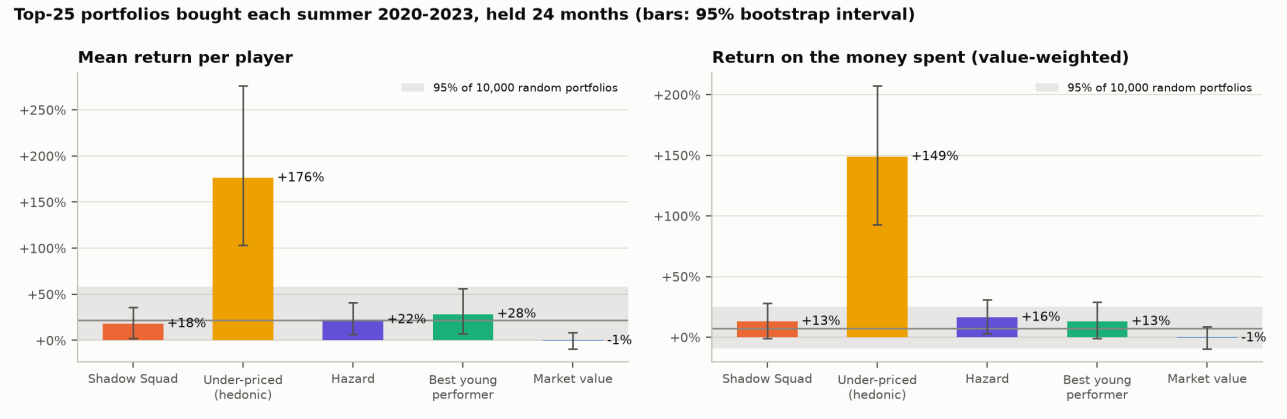


Figure 16. Portfolio returns against the distribution of 10,000 random portfolios (grey band).

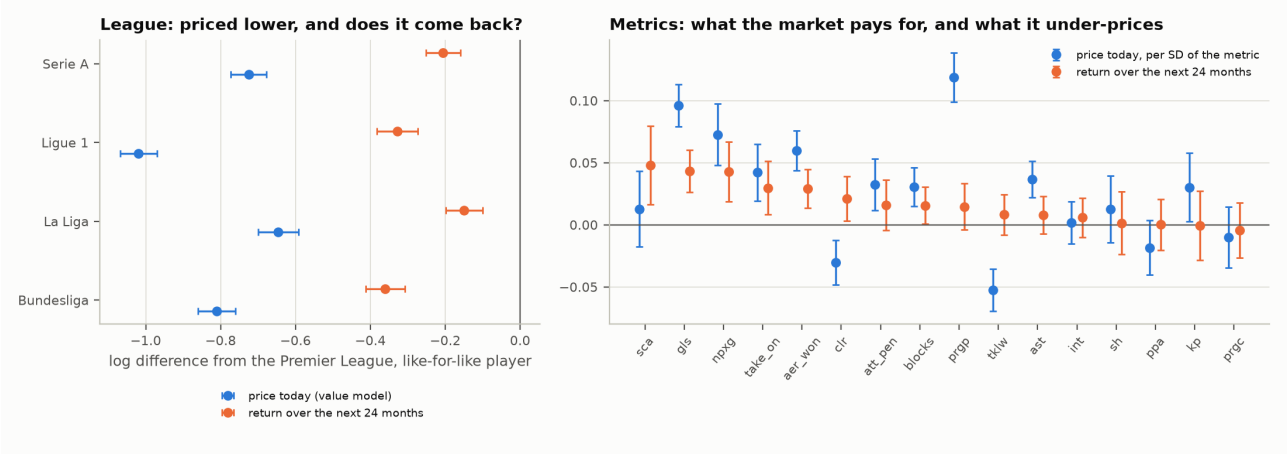


Figure 17. Hedonic model. Blue: what the market pays for today. Orange: what predicts the next two years of return, holding today's price fixed.

7B. Sliding doors: does moving to an elite club help a young player?

Treated: a feeder (24 or under, non-elite club, 900 minutes in season t) who makes a permanent move to a club that is elite in t+1 (190 players). Control: a feeder with no move of any kind that year (1,182). Twins are proposed by the Spark LSH index (same fine position and season; nearest in age, style-neutral profile, minutes, club table position and market value), then matched one to one on a propensity score. The matching setting was chosen on covariate balance alone, before any outcome was looked at: 25 LSH twins per treated player and a caliper of 0.2 SD, giving 142 pairs. The worst standardised mean difference falls from 0.41 to 0.12; 2 of 21 covariates stay just above the 0.1 target (log value, blocks). Outcomes are changes from season t to t+2: share of league minutes, and the composite index of *style-neutral* per-90 output, which is what makes a season at Arsenal comparable with one at Brentford.

Outcome, change t to t+2	Pairs	Moved to elite	Twin who stayed	Difference [95% CI]	p	E-value
Share of league minutes	142	-18.6 pts	-20.4 pts	+1.8 pts [-4.3 to +7.9]	0.57	1.29
Style-neutral output index (0-100)	79	-1.0	-1.3	-0.05 [-2.8 to +2.9]	0.97	1.07
Minutes, causal forest: 21 or under	91			-1.0 pts [-9.1 to +7.1]	0.81	
Minutes, causal forest: 22	25			-1.0 pts [-9.4 to +7.4]	0.81	
Minutes, causal forest: 23-24	74			-0.5 pts [-8.8 to +7.7]	0.90	

Table 7. Effect of moving to an elite club. Rows 1-2: matched pairs, bootstrap intervals, sign-flip p. Rows 3-5: CausalForestDML on all 1,372 feeders (pairs column = treated players in the group). E-value: how strongly an unmeasured confounder would have to be tied to both the move and the outcome to produce the point estimate.

HEADLINE

Moving to an elite club at 24 or under changes a player's share of league minutes by +1.8 points (95% CI -4.3 to +7.9) and his style-neutral output by -0.05 index points (-2.8 to +2.9) against a twin who stayed. Neither is distinguishable from zero, and the effect does not flip by age: -1.0 points at 21 or under, -0.5 at 23-24.

Both groups lose minutes over two years (-19 and -20 points), which is regression to the mean: everyone was picked for having played 900 minutes. The common story that a big move costs a young player his playing time is not in this data, and neither is a gain in output. The causal forest finds no subgroup with an effect; its estimates by position are DF -0.1, MF -1.3, FW -1.1 points, all with intervals about ±8 points wide. Among matched pairs the estimate is the same whether the buying club is more or less dominant than the median buyer (+3.6 and +0.1 points).

Limitations. A null with an interval of about ±6 points rules out a large effect, not a small one. Output is observed only for players with 900 minutes in t+2 (56% of pairs), so the output comparison is among those who kept playing. Clubs choose whom to buy on things the data does not hold (injuries, character, contract length); the E-values near 1 say that even a weak hidden factor could move estimates this small in either direction. Balance is close to, not inside, the 0.1 target.

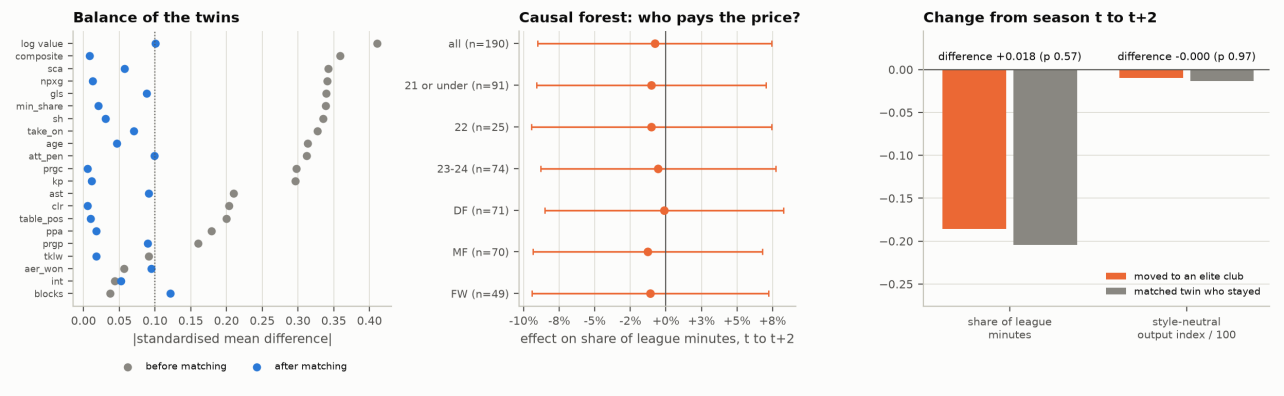


Figure 18. Left: covariate balance before and after matching. Centre: causal-forest effect on minutes by age and position. Right: change in each outcome for movers and their twins.

7C. Red-team the scout: can a player game the model?

Four of the sixteen metrics are volume a player controls with little regard to quality: shots, take-on attempts, clearances, progressive carries. The other twelve need an outcome or a team-mate. For every player ranked 11 to 20 in 150 unseen replacement queries (1,500 players), a greedy search raises his cheap metrics in steps of 5% of the position's 5th-95th percentile range, never above the 95th percentile, and re-scores him through the whole pipeline: style-neutral value, standardisation, translated similarity, trajectory, composite index and hazard. He games the model if he enters the top 10.

	Engine	Hardened (ignores cheap metrics)
Gameability: near-miss players who can pad into the top 10	29.7% [27, 32]	0% (by construction)
Median padding needed	1 step: 5% of the position's range	not applicable
Most used metric, median amount added	shots: +0.09 per 90 (+9%)	not applicable
Recall@10 on 91 unseen replacements	31%	36% (change -1 to +13 points)
Recall@25, median rank	60%, 18	60%, 18

Table 8. Gameability before and after hardening, and what hardening costs.

HEADLINE

Adding 0.09 shots per 90 is enough to lift a near-miss player into the shortlist; 29.7% of near-miss players can do it with cheap stats alone. The hardened model cuts this to zero and recall@10 does not fall (31% to 36%).

The amounts are small because the margins are: ranks 10 and 11 are separated by very little score, so one step is usually enough. The hardened model computes similarities and trajectory on the twelve costly metrics, uses the mean costly percentile as its composite index, and refits the hazard without cheap metrics. Padding then changes nothing it sees. That it loses no recall says the cheap metrics were carrying little real signal.

Limitations. The cheap and costly tags are a judgement, not a measurement. The search changes one season's numbers and holds archetype, club pipeline and market value fixed; a real padding campaign would also move the crowd's valuation. Zero gameability is by construction and only covers the four tagged metrics: a player can still improve costly metrics, which is the point. Recall changes are measured on 91 events and their interval is wide.

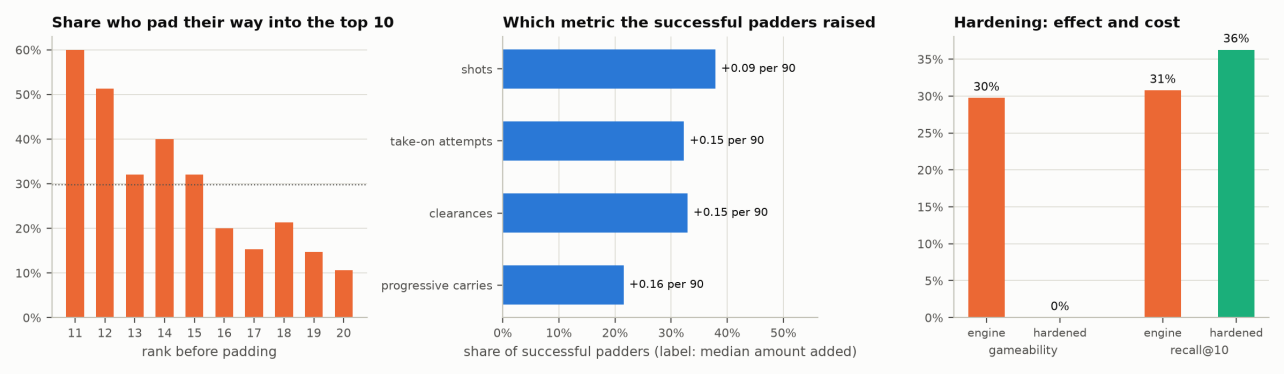


Figure 19. Left: share of players who can pad into the top 10, by starting rank. Centre: which cheap metric the successful ones raised. Right: gameability and recall@10 before and after hardening.

7D. Budget squad rebuild (integer programming) was a stretch goal and was not built.

8 Model predictions and case studies

Dashboard · SHAP · Translated radar

The screenshots are from the Streamlit app (*app/shadow_squad.py*), which reads the Parquet lake: choose a league, an elite club, a season and a departing starter. It shows the ranked list, what the club really did, a radar (starter's actual numbers, candidate raw, candidate translated into the club), a SHAP waterfall for the chosen candidate, and market values. The cases were picked by rule, not by hand: the three unseen events with the best rank for the real signing (ties broken by the larger fee), and the unseen event where a real signing was ranked worst.

Across all unseen events the scores that move the ranking most are value gap, age, market value, hazard (mean absolute SHAP).

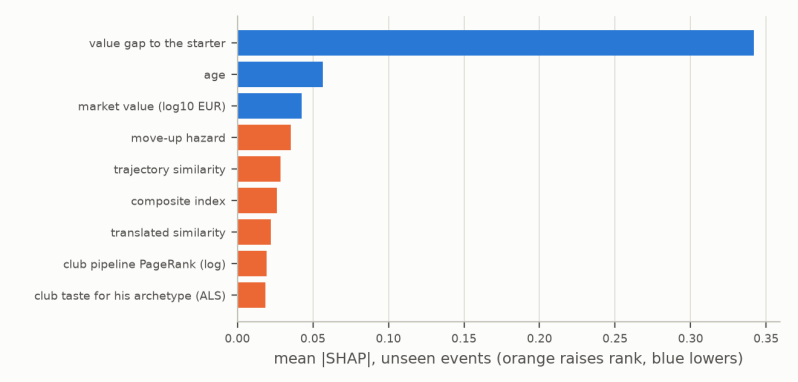


Figure 20. Mean absolute SHAP value of each score on unseen events.

Manchester City 2024-25: Rico Lewis (FB) replaced by Rayan Aït-Nouri from Wolves (EUR 36.8m). The engine ranked him 1 of 103 candidates. His strongest scores were closeness in value to the starter (better than 99% of the pool), move-up hazard (better than 99% of the pool), market value (better than 96% of the pool). The waterfall shows which scores carried him there, and the radar how his numbers sit against the starter's once translated into the club.

rank	player	club	age	value (EUR)	score	translated sim.	trajectory sim.	hazard
1	Rayan Aït-Nouri	Wolves	23	35,000,000	0.88	0.01	-0.09	40%
2	Antonee Robinson	Fulham	26	35,000,000	0.45	-0.46	-0.34	15%
3	Milos Kerkez	Bournemouth	20	45,000,000	0.44	0.23	0.06	23%
4	Miguel Gutiérrez	Girona	23	25,000,000	0.43	0.24	0.21	18%
5	Tyrick Mitchell	Crystal Palace	24	25,000,000	0.42	-0.08	-0.03	16%
6	Lorenz Assignon	Rennes	24	9,000,000	0.41	-0.07	-0.17	27%
7	Tom Rothe	Union Berlin	19	10,000,000	0.40	0.22	0.26	16%
8	Lutsharel Geertruida	RB Leipzig	24	27,000,000	0.40	0.23	0.23	5%
9	Nuno Tavares	Lazio	24	25,000,000	0.39	-0.17	-0.11	16%
10	Guela Doué	Strasbourg	21	12,000,000	0.38	0.22	0.19	20%

Figure 21a. Ranked list for this event (top 10), from the dashboard.

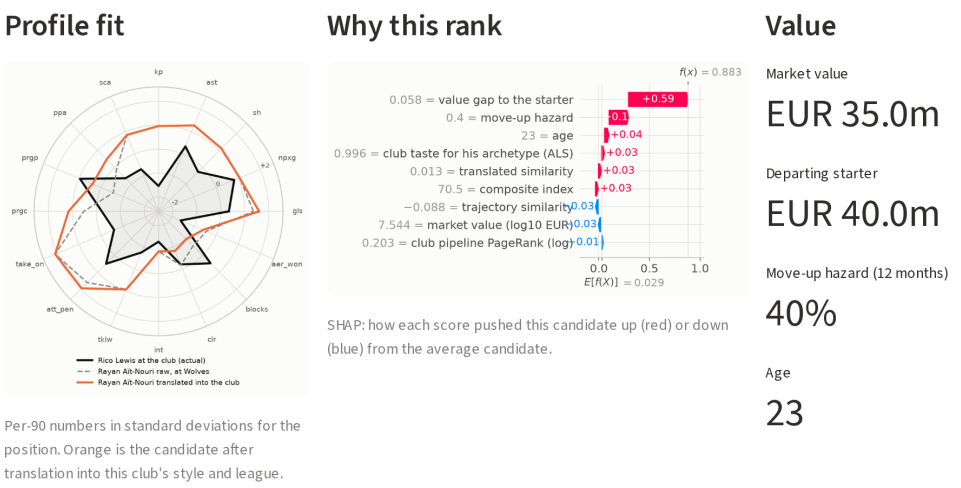


Figure 21b. Rayan Aït-Nouri: translated radar, SHAP waterfall and value panel.

Rennes 2022-23: Flavien Tait (CM) replaced by Enzo Le Fée from FC Lorient (EUR 20.0m). The engine ranked him 1 of 89 candidates. His strongest scores were composite index (better than 99% of the pool), move-up hazard (better than 98% of the pool), trajectory similarity (better than 92% of the pool). The waterfall shows which scores carried him there, and the radar how his numbers sit against the starter's once translated into the club.

rank	player	club	age	value (EUR)	score	translated sim.	trajectory sim.	hazard
1	Enzo Le Fée	Lorient	22	15,000,000	0.62	0.67	0.53	33%
2	Jean-Ricner Bellegarde	Strasbourg	24	10,000,000	0.54	0.82	0.44	29%
3	Habib Diarra	Strasbourg	18	4,000,000	0.44	0.65	0.59	26%
4	Arne Maier	Augsburg	23	5,500,000	0.40	0.27	0.11	14%
5	Kiernan Dewsbury-Hall	Leicester City	23	25,000,000	0.39	0.79	0.63	16%
6	Khéphren Thuram	Nice	21	32,000,000	0.38	0.75	0.29	33%
7	Azzedine Ounahi	Angers	22	15,000,000	0.36	0.78	0.68	17%
8	Ilaix Moriba	Valencia	19	6,000,000	0.33	0.37	0.20	11%
9	Han-Noah Massengo	Auxerre	21	4,000,000	0.33	0.51	0.44	9%
10	Kike Pérez	Valladolid	25	4,500,000	0.33	0.72	0.12	13%

Figure 22a. Ranked list for this event (top 10), from the dashboard.

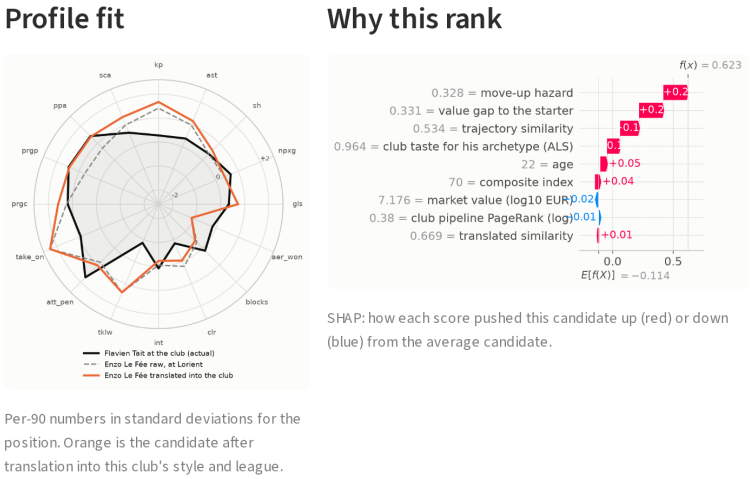


Figure 22b. Enzo Le Fée: translated radar, SHAP waterfall and value panel.

Marseille 2021-22: Pol Lirola (FB) replaced by Issa Kaboré from Man City (free or undisclosed fee). The engine ranked him 1 of 111 candidates. His strongest scores were trajectory similarity (better than 97% of the pool), translated similarity (better than 93% of the pool), move-up hazard (better than 80% of the pool). The waterfall shows which scores carried him there, and the radar how his numbers sit against the starter's once translated into the club.

rank	player	club	age	value (EUR)	score	translated sim.	trajectory sim.	hazard
1	Issa Kaboré	Troyes	20	4,000,000	0.54	0.65	0.63	18%
2	Destiny Udogie	Udinese	18	9,000,000	0.44	0.50	0.43	32%
3	Manu Sánchez	Osasuna	20	10,000,000	0.43	0.24	0.16	17%
4	David Raum	Hoffenheim	23	17,000,000	0.42	0.68	0.63	20%
5	Quentin Merlin	Nantes	19	5,000,000	0.35	0.28	0.17	17%
6	Akim Zedadka	Clermont Foot	26	4,000,000	0.31	0.63	0.56	17%
7	Fran Garcia	Rayo Vallecano	21	5,000,000	0.31	0.41	0.29	22%
8	Mathías Olivera	Getafe	23	15,000,000	0.30	0.58	0.10	32%
9	Yvann Macon	Saint-Étienne	22	4,500,000	0.29	0.24	0.12	9%
10	Brandon Williams	Norwich City	20	10,000,000	0.26	0.12	0.08	12%

Figure 23a. Ranked list for this event (top 10), from the dashboard.

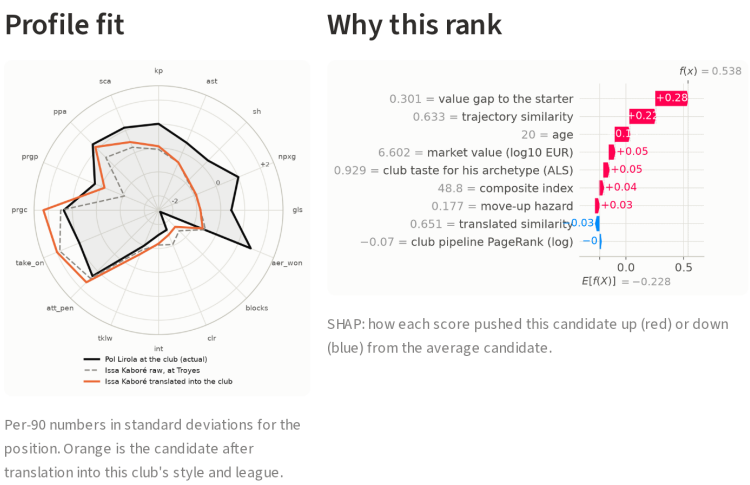


Figure 23b. Issa Kaboré: translated radar, SHAP waterfall and value panel.

A miss

Chelsea 2023-24: Thiago Silva (CB) replaced by Tosin Adarabioyo from Fulham (free or undisclosed fee). The engine ranked him 131 of 141, almost last. His weakest scores were closeness in value to the starter (better than 13% of the pool), club pipeline rank (better than 19% of the pool), club taste (better than 28% of the pool). He was valued at EUR 20.0m and the starter he replaced at EUR 1.0m. On the scores the engine uses he looked nothing like the man he replaced, so nothing pointed to him; whatever the club saw is not in the data.

rank	player	club	age	value (EUR)	score	translated sim.	trajectory sim.	hazard
1	Reece Burke	Luton Town	26	1,800,000	0.34	0.18	0.19	5%
2	Matej Maglica	Darmstadt 98	24	2,000,000	0.33	0.20	0.29	6%
3	Souleymane Touré	Lorient	20	8,000,000	0.31	-0.07	-0.03	33%
4	Fali Candé	Metz	25	2,500,000	0.26	0.11	-0.07	8%
5	Chumi	Almería	24	2,000,000	0.26	0.16	-0.25	7%
6	Thomas Kristensen	Udinese	21	4,500,000	0.26	-0.45	-0.29	9%
7	Abdul Mumin	Rayo Vallecano	25	2,000,000	0.25	0.44	0.05	5%
8	Yerson Mosquera	Villarreal	22	3,500,000	0.24	0.05	0.10	18%
9	Cenk Özkacar	Valencia	22	2,000,000	0.23	-0.30	-0.34	6%
10	Christoph Klarer	Darmstadt 98	23	3,000,000	0.23	0.21	0.27	9%

Figure 24a. Ranked list for this event (top 10), from the dashboard.

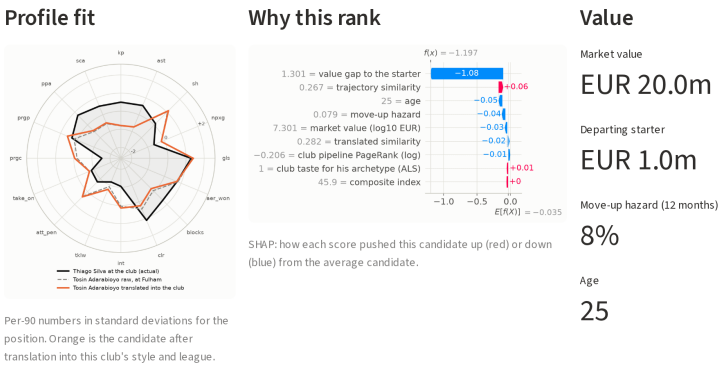


Figure 24b. Tosin Adarabioyo: translated radar, SHAP waterfall and value panel.

9 Limitations and conclusion

Honest reading

- **Coverage.** 76% of real elite replacement signings were outside the candidate pool, most from leagues the data does not cover. The all-signings test only includes players who had 900 minutes in the five leagues the season before.
- **Sample size.** 128 in-pool elite replacements in eight seasons, 91 of them unseen; intervals on that task are wide, a single model moves by several points with its seed, and the order of the top methods is not reliable. The all-signings test (1,029 queries) is where differences are measured with confidence.
- **Transfer records.** 4.0% of feeder-seasons belong to players with no transfer history even after inferring moves from minutes, and are labelled "no move". Loans are inferred from move patterns, not from a loan flag. 49% of signings have no fee above zero recorded.
- **Market value.** Transfermarkt values are crowd estimates; used strictly before the window, they still encode what the crowd expects.
- **Local Spark.** Everything runs in local mode on one machine, on data far smaller than Spark is for.

Reframing the task as "whom does the club sign" gives a test the mid-sem lacked: a real decision by a real club. On 1,029 unseen signings the engine is clearly better than every baseline, with 3.1 times the top-10 recall of market value. On elite replacements it is level with market value and the hazard score, and the fee split on the large set explains why: clubs paying big fees buy the players everyone already rates, and the engine's edge is below the median fee. The market itself can be read: players priced below what their profile predicts returned +176% in two years. Moving to an elite club neither cost nor helped a young player on average. The ranker can be gamed with cheap stats, and ignoring them costs nothing. Style translation, ALS club taste and the hidden-gem flag failed under walk-forward fitting.